

For the use of a Registered Medical Practitioner Only

PRESCRIBING INFORMATION

DEMEALRAN TABLETS

(Donepezil hydrochloride 5/10 mg tablets)

COMPOSITION

Each film-coated tablet contains:

Donepezil hydrochloride.....5/10 mg

Excipients: Lactose monohydrate, maize starch, hydroxypropyl cellulose, purified water, microcrystalline cellulose, sodium starch glycolate, magnesium stearate, opadry 02b52480 yellow [hypromellose 5 cp, titanium dioxide (E171), macrogol (400), talc, iron oxide yellow (E172)].

PRODUCT DESCRIPTION

DEMEALRAN TABLETS 5 mg

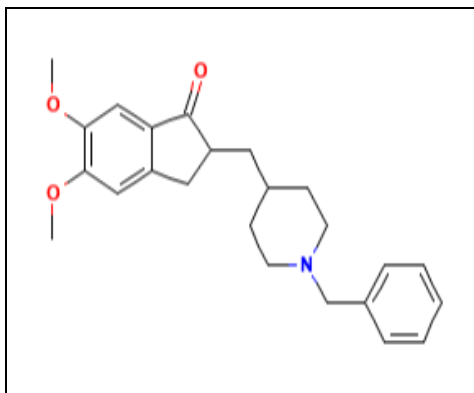
Yellow coloured, circular, biconvex film coated tablets debossed with 'RC25' on one side.

DEMEALRAN TABLETS 10 mg

Yellow coloured, capsule shaped, film coated tablets debossed with 'RC' & '26' on either side of the breakline on one side and a breakline on the other side.

DESCRIPTION

DEMEALRAN TABLET contains an active substance donepezil. Donepezil is an oral acetylcholinesterase inhibitor used for therapy of Alzheimer disease.. It is chemically known as 2-((1-Benzylpiperidin-4-yl)methyl)-5,6-dimethoxy-2,3-dihydro-1H-inden-1-one. Molecular formula is $C_{24}H_{29}NO_3$ and molecular weight is 379.5 g/mol. The structural formula of donepezil is as follows:



STRUCTURAL FORMULA OF DONEPEZIL

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamic properties

Donepezil hydrochloride is a specific and reversible inhibitor of acetylcholinesterase, the predominant cholinesterase in the brain. Donepezil hydrochloride is over 1000 times more potent an inhibitor of this enzyme *in vitro* than of butyrylcholinesterase, an enzyme which is present mainly outside the central nervous system.

Pharmacokinetics

Absorption

Maximum plasma levels have been reported to be reached approximately 3 to 4 hours after oral administration. Plasma concentrations and area under the curve rise in proportion to the dose. The terminal disposition half-life has been reported to be approximately 70 hours, thus, administration of multiple single-daily doses results in gradual approach to steady-state. Approximate steady-state is achieved within 3 weeks after initiation of therapy. Once at steady-state, plasma donepezil hydrochloride concentrations and the related pharmacodynamic activity show little variability over the course of the day.

It has been reported that food did not affect the absorption of donepezil hydrochloride.

Distribution

Donepezil hydrochloride has been reported to be approximately 95% bound to human plasma proteins. The plasma protein binding of the active metabolite 6-O-desmethyl donepezil is not known. The distribution of donepezil hydrochloride in various body tissues has not been definitively reported.

Metabolism/Excretion

Donepezil hydrochloride is both excreted in the urine intact and metabolized by the cytochrome P450 system to multiple metabolites, not all of which have been identified. Following administration of a single 5 mg dose of ¹⁴C-labeled donepezil hydrochloride, plasma radioactivity, expressed as a percent of the administered dose, has been reported to be present primarily as intact donepezil hydrochloride (30%), 6-O-desmethyl donepezil (11% - only metabolite that exhibits activity similar to donepezil hydrochloride), donepezil-cis-N-oxide (9%), 5-O-desmethyl donepezil (7%) and the glucuronide conjugate of 5-O-desmethyl donepezil (3%).

Approximately 57% of the total administered radioactivity has been reported to be recovered from the urine (17% as unchanged donepezil), and 14.5% recovered from the faeces, suggesting biotransformation and urinary excretion as the primary routes of elimination. There is no reported evidence to suggest enterohepatic recirculation of donepezil hydrochloride and/or any of its metabolites.

Plasma donepezil concentrations decline with a half-life of approximately 70 hours.

Sex, race and smoking history have no clinically significant influence on plasma concentrations of donepezil hydrochloride. The pharmacokinetics of donepezil has not been formally reported in healthy

elderly subjects, or in Alzheimer's or vascular dementia patients. However, mean plasma levels in patients closely agreed with those of young healthy volunteers.

Patients with mild to moderate hepatic impairment has been reported to increase donepezil steady state concentrations; mean AUC by 48% and mean $C_{\max}$ by 39% (see **DOSE AND METHOD OF ADMINISTRATION**).

INDICATIONS

DEMEALRAN TABLET is indicated for the symptomatic treatment of mild, moderate, and severe dementia in Alzheimer's disease.

DOSE AND METHOD OF ADMINISTRATION

Adults/Elderly

Mild to Moderate Alzheimer's disease

Treatment is initiated at 5 mg/day (once-a-day dosing). Donepezil should be taken orally, in the evening, just prior to retiring. The 5 mg/day dose should be maintained for at least one month in order to allow the earliest clinical responses to treatment to be assessed and to allow steady-state concentrations of donepezil hydrochloride to be achieved. Following a one-month clinical assessment of treatment at 5 mg/day, the dose of donepezil can be increased to 10 mg/day (once-a-day dosing). The maximum recommended daily dose is 10 mg.

Severe Alzheimer's disease

Recommended dose is 10 mg administered once daily.

The recommended starting dose of donepezil is 5 mg once daily. It has been reported that in mild to moderate Alzheimer's disease the 10 mg dose, with one week titration, is likely to be associated with a higher incidence of cholinergic adverse events compared to the 5 mg dose. During a 6 week titration, it has been reported that the type of frequency of these same adverse events were similar between the 5 mg and 10mg dose groups. Therefore, because donepezil steady state is achieved about 15 days after it is started and because the incidence of untoward effects may be influenced by the rate of dose escalation, a dose of 10 mg should not be administered until patients have been on a daily dose of 5 mg for 4 to 6 weeks.

General Consideration

Treatment should be initiated and supervised by a physician experienced in the diagnosis and treatment of Alzheimer's dementia. Diagnosis should be made according to accepted guidelines (e.g. DSM IV, ICD 10). Therapy with donepezil should only be started if a caregiver is available who will regularly monitor drug intake for the patient. Maintenance treatment can be continued for as long as a therapeutic benefit for the patient exists. Therefore, the clinical benefit of donepezil should be reassessed on a regular basis. Discontinuation should be considered when evidence of a therapeutic *effect* is no longer present. Individual response to donepezil cannot be predicted.

Upon discontinuation of treatment, a gradual abatement of the beneficial effects of donepezil has been reported. There is no reported evidence of a rebound effect after abrupt discontinuation of therapy.

Renal impairment

A similar dose schedule can be followed for patients with renal impairment as clearance of donepezil hydrochloride is not affected by this condition.

Hepatic impairment

Due to possible increased exposure in mild to moderate hepatic impairment, dose escalation should be performed according to individual tolerability. No data have been reported for patients with severe hepatic impairment.

Children:

Donepezil is not recommended for use in children.

CONTRAINDICATIONS

DEMEALRAN TABLETS is contraindicated in patients with a known hypersensitivity to donepezil hydrochloride, piperidine derivatives, or to any excipients used in the formulation.

WARNINGS AND PRECAUTIONS

The use of donepezil in patients with severe dementia, other types of dementia or other types of memory impairment (e.g., age-related cognitive decline), has not been reported.

Anaesthesia

Donepezil, as a cholinesterase inhibitor, is likely to exaggerate succinylcholine-type muscle relaxation during anaesthesia.

Cardiovascular Conditions

Because of their pharmacological action, cholinesterase inhibitors may have vagotonic effects on heart rate (e.g. bradycardia). The potential for this action may be particularly important to patients with "sick sinus syndrome" or other supraventricular cardiac conduction conditions, such as sinoatrial or atrioventricular block.

Syncope and seizures can occur with donepezil. In investigating such patients the possibility of heart block or long sinus pauses should be considered.

QTc interval prolongation and Torsade de Pointes can occur with donepezil (see **DRUG INTERACTIONS and SIDE EFFECTS/UNDESIRABLE EFFECTS**). Caution is advised in patients with pre-existing or family history of QTc prolongation, in patients treated with drugs affecting the QTc

interval, or in patients with relevant pre-existing cardiac disease (e.g. uncompensated heart failure, recent myocardial infarction, bradyarrhythmias), or electrolyte disturbances (hypokalaemia, hypomagnesaemia). Clinical monitoring (ECG) may be required.

Gastrointestinal Conditions

Patients at increased risk for developing ulcers, e.g., those with a history of ulcer disease or those receiving concurrent nonsteroidal anti-inflammatory drugs (NSAIDs), should be monitored for symptoms. However, Donepezil has not been reported to increase either peptic ulcer disease or gastrointestinal bleeding.

Genitourinary

Although not reported in clinical studies of donepezil, cholinomimetics may cause bladder outflow obstruction.

Neurological Conditions

Seizures: Cholinomimetics are believed to have some potential to cause generalised convulsions. However, seizure activity may also be a manifestation of Alzheimer's disease.

Cholinomimetics may have the potential to exacerbate or induce extrapyramidal symptoms.

Neuroleptic Malignant Syndrome (NMS)

NMS, a potentially life-threatening condition characterised by hyperthermia, muscle rigidity, autonomic instability, altered consciousness and elevated serum creatine phosphokinase levels, has been reported to occur very rarely in association with donepezil, particularly in patients also receiving concomitant antipsychotics. Additional signs may include myoglobinuria (rhabdomyolysis) and acute renal failure. If a patient develops signs and symptoms indicative of NMS, or presents with unexplained high fever without additional clinical manifestations of NMS, treatment should be discontinued.

Pulmonary Conditions

Because of their cholinomimetic actions, cholinesterase inhibitors should be prescribed with care to patients with a history of asthma or obstructive pulmonary disease.

The administration of donepezil concomitantly with other inhibitors of acetylcholinesterase, agonists or antagonists of the cholinergic system should be avoided.

Severe Hepatic impairment

No data has been reported for patients with severe hepatic impairment.

Mortality in Vascular Dementia In the first reported study, the mortality rates were 1.0% on donepezil hydrochloride 5 mg, 2.4% on donepezil hydrochloride 10 mg and 3.5% on placebo. In the second reported study, the mortality rates were 1.9% on donepezil hydrochloride 5 mg, 1.4% on donepezil hydrochloride 10 mg and 0.5% on placebo. In the third reported study, the mortality rates were 1.7% on donepezil hydrochloride 5 mg and 0% on placebo. The mortality rate for the three VaD reported studies combined in the donepezil hydrochloride group (1.7%) was numerically higher than in the placebo group (1.1%), however, this difference has not been reported to be statistically significant.

The majority of deaths in patients taking either donepezil hydrochloride or placebo appear to result from various vascular related causes, which could be expected in this elderly population with underlying vascular disease. An analysis of all serious non-fatal and fatal vascular events reported no difference in the rate of occurrence in the donepezil hydrochloride group relative to placebo.

Effects on Ability to Drive and Use Machines

Donepezil has minor or moderate influence on the ability to drive and use machines.

Dementia may cause impairment of driving performance or compromise the ability to use machinery. Furthermore, donepezil can induce fatigue, dizziness and muscle cramps, mainly when initiating or increasing the dose. The treating physician should routinely evaluate the ability of patients on donepezil to continue driving or operating complex machines.

DRUG INTERACTIONS

Donepezil hydrochloride and/or any of its metabolites does not inhibit the metabolism of theophylline, warfarin, cimetidine or digoxin in humans. The metabolism of donepezil hydrochloride is not affected by concurrent administration of digoxin or cimetidine.

Cytochrome P450 isoenzymes 3A4 and to a minor extent 2D6 are involved in the metabolism of donepezil *in vitro*. Ketoconazole and quinidine, inhibitors of CYP3A4 and 2D6 respectively, inhibit donepezil metabolism *in vitro*. Therefore, these and other CYP3A4 inhibitors, such as itraconazole and erythromycin, and CYP2D6 inhibitors, such as fluoxetine, could inhibit the metabolism of donepezil. In healthy volunteers, ketoconazole increased mean donepezil concentrations by about 30%.

Enzyme inducers, such as rifampicin, phenytoin, carbamazepine and alcohol may reduce the levels of donepezil. Since the magnitude of an inhibiting or inducing *effect* is unknown, such drug combinations should be used with care. Donepezil hydrochloride has the potential to interfere with medications having anticholinergic activity. There is also the potential for synergistic activity with concomitant treatment involving medications such as succinylcholine, other neuro-muscular blocking agents or cholinergic agonists or beta blocking agents that have *effects* on cardiac conduction.

Cases of QTc interval prolongation and Torsade de Pointes have been reported for donepezil. Caution is advised when donepezil is used in combination with other medicinal products known to prolong the QTc interval and clinical monitoring (ECG) may be required. Examples include:

- Class IA antiarrhythmics (e.g. quinidine)
- Class III antiarrhythmics (e.g. amiodarone, sotalol)
- Certain antidepressants (e.g. citalopram, escitalopram, amitriptyline)
- Other antipsychotics (e.g. phenothiazine derivatives, sertindole, pimozide, ziprasidone)
- Certain antibiotics (e.g. clarithromycin, erythromycin, levofloxacin, moxifloxacin)

UNDESIRABLE EFFECTS

The most common adverse events have been reported to be diarrhoea, muscle cramps, fatigue, nausea, vomiting and insomnia.

Adverse reactions reported as more than an isolated case are listed below, by system organ class and by frequency. Frequencies are defined as: very common ($\geq 1/10$) common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $1/100$), rare ($\geq 1/10,000$ to $1/1,000$); very rare ($< 1/10,000$) and not known (cannot be estimated from reported data).

System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare	Not Known
Infections and infestations		Common cold				
Metabolism and nutrition disorders		Anorexia				
Psychiatric disorders		Hallucinations** Agitation** Aggressive behaviour** Abnormal dreams and Nightmares**				Libido increased, Hypersexuality
Nervous system disorders		Syncope* Dizziness Insomnia	Seizure*	Extrapyramidal symptoms	Neuroleptic malignant syndrome	Pleurothotonus (Pisa syndrome)
Cardiac disorders			Bradycardia	Sino-atrial block Atrioventricular block		Polymorphic ventricular tachycardia including Torsade de Pointes Electrocardiogram QT interval prolonged
Gastrointestinal disorders	Diarrhoea Nausea	Vomiting Abdominal disturbance	Gastrointestinal haemorrhage Gastric and duodenal ulcers Salivary hypersecretion			
Hepato-biliary disorders				Liver dysfunction including hepatitis***		
Skin and		Rash				

subcutaneous tissue disorders		Pruritis				
Musculoskeletal, connective tissue and bone disorders		Muscle cramps			Rhabdomyolysis ****	
Renal and urinary disorders		Urinary incontinence				
General disorders and administration site conditions	Headache	Fatigue Pain				
Investigations			Minor increase in serum concentration of muscle creatine kinase			
Injury and poisoning		Accidents including falls				

* In investigating patients for syncope or seizure the possibility of heart block or long sinus pauses should be considered (see **WARNINGS AND PRECAUTIONS**).

** Reports of hallucinations, abnormal dreams, nightmares, agitation and aggressive behaviour have been reported to be resolved on dose-reduction or discontinuation of treatment.

*** In cases of unexplained liver dysfunction, withdrawal of donepezil should be considered.

**** Rhabdomyolysis has been reported to occur independently of Neuroleptic Malignant syndrome and in close temporal association with donepezil initiation or dose increase.

Reported post-marketing experience

Hallucinations, agitation, aggressive behavior, seizure, hepatitis, gastric ulcer, duodenal ulcer, and gastrointestinal hemorrhage has been reported with donepezil.

USE IN SPECIAL POPULATIONS

Pregnancy:

There are no adequate data reported from the use of donepezil in pregnant women.

Teratogenic effect has not been reported in animals but pre and post natal toxicity have been reported. The potential risk for humans has not been reported.

Donepezil should not be used during pregnancy unless clearly necessary.

Lactation

Donepezil is excreted in the milk of rat. It is not known whether donepezil hydrochloride is excreted in human breast milk and there are no studies reported in lactating women. Therefore, women on donepezil should not breast feed.

OVERDOSE

The estimated median lethal dose of donepezil hydrochloride following administration of a single oral dose in mice and rats in 45 and 32 mg/kg, respectively, or approximately 225 and 160 times the maximum recommended human dose of 10 mg per day. Dose-related signs of cholinergic stimulation have been reported in animals and included reduced spontaneous movement, prone position, staggering gait, lacrimation, clonic convulsions, depressed respiration, salivation, miosis, fasciculation and lower body surface temperature.

Over dosage with cholinesterase inhibitors can result in cholinergic crisis characterized by severe nausea, vomiting, salivation, sweating, bradycardia, hypotension, respiratory depression, collapse and convulsions. Increasing muscle weakness is a possibility and may result in death if respiratory muscles are involved.

As in any case of overdose, general supportive measures should be utilised. Tertiary anticholinergics such as atropine may be used as an antidote for donepezil over dosage. Intravenous atropine sulphate titrated to effect is recommended: an initial dose of 1.0 to 2.0 mg IV with subsequent doses based upon clinical response. Atypical responses in blood pressure and heart rate have been reported with other cholinomimetics when co-administered with quaternary anticholinergics such as glycopyrrolate. It is not known whether donepezil hydrochloride and/or its metabolites can be removed by dialysis (hemodialysis, peritoneal dialysis, or hemofiltration).

Pack Style:

Blister pack of 7's, Box of 4 x 7's

Blister pack of 10's, Box of 3 x 10's

STORAGE

Store at a controlled temperature below 30°C

Manufacturer

Sun Pharmaceutical Industries Limited.,
Paonta Sahib, Dist. Sirmour
Himachal Pradesh 173 025
India

Product Registration Holder

RANBAXY (MALAYSIA) SDN. BHD.
Lot 23, Bakar Arang Industrial Estate
08000 Sungai Petani,
Kedah, Malaysia

Information compiled in January 2023

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